

SQL

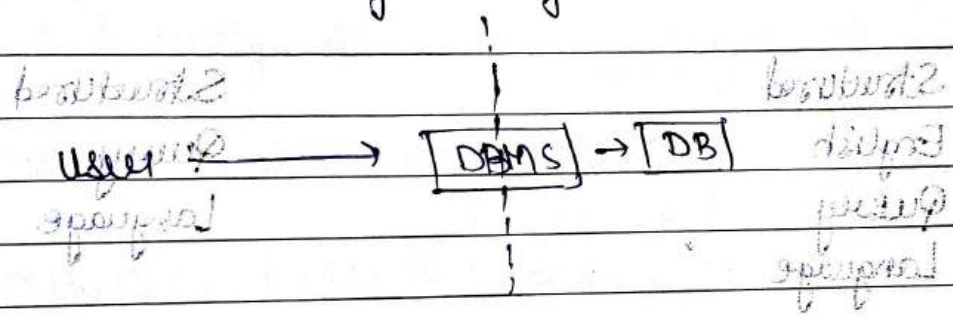
structured query language

computer language for storing & retrieving data

① Database :-

② Database is collection of data in a format that can be easily accessed (Digital).

A software application used to manage our DB is called DBMS (Database Management System).



→ Type of database :-

Relational

Data stored in tables

Ex- MySQL
SQL Server

Non-relational (NoSQL)

Data not stored in tables

Ex- MongoDB

What is SQL

SQL → Structured Query Language.

SQL is a programming language used to interact with

relational database.

Note ① It is used to perform CRUD operations:-

C → Create.

R → Read.

U → Update.

D → Delete.

SEQUEL

Structured

Structured

English

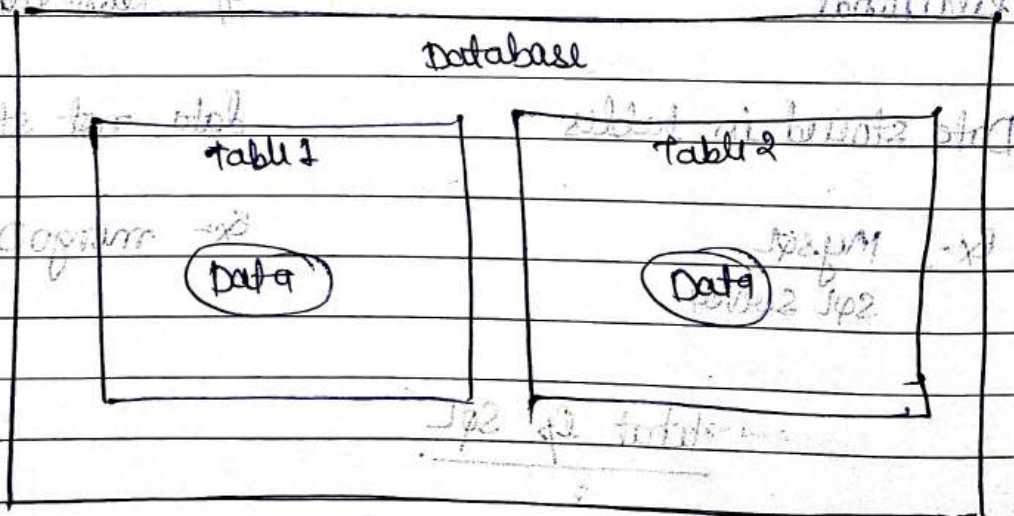
Query

Language

Query

Language

② Database structure:-



③ What is table:-

Roll no.	SNo.	Name	Age	MM
10	1	XYZ	19	20
20	2	ABC	18	24

column 1
column 2

① Spl Datatypes

② Signed & unsigned

TINYINT UNSIGNED (0 to 255)

TINYINT (-128 to 127)

By using unsigned we can increase the size in positive.

③ Types of Spl commands

1. DDL (Data Definition language)

create
alter
rename
truncate & drop

2. DQL (Data Query language)

select

3. DMC (Data Manipulation language) :-
 insert
 update
 delete

4. DCL (Data Control language) :-
 grant & revoke permission to users

5. TCL (Transaction Control language) :-
 start transaction
 commit
 rollback

6. Database related queries

create {
 CREATE DATABASE db-name;
 CREATE DATABASE IF NOT EXISTS db-name;

drop {
 DROP DATABASE db-name;
 DROP DATABASE IF EXISTS db-name;

7. To see all database
 SHOW DATABASES;

① To see all tables

no. in (cancel) SHOW TABLES; cancel a 2nd request A
not wait. post processing at at before with ident.

⑥ To create table: $\rightarrow$ "create table" statement is used

1. CREATE TABLE table name (no. of request)

column_name1	datatypes	constraints
column_name2	datatypes	constraints

prob of value pfings of been no strindens 100

① Table related queries

• gati / (Selecio furtivo alla colombaria) 100% 70%

SELECT * (FROM table name).

2. ~~try to insert~~ insert some thing in table. SUPPLEMENT

INSERT INTO table_name VALUES (-----)

page # 11

keys:- both entities traversed
.. and not removed & still

1. Primary key :-

② It is a column (or set of columns) in a table that uniquely identifies each row. (a unique id)

There is only 1 primary key it should be not null.

2. Foreign key:-

oldot llo 12 01 ()

① A foreign key is a column (or set of columns) in a table that refers to the primary key. There can be multiple foreign keys in table.

Foreign keys can have duplicate & Null values.

② Constraints:- *sql table kore - mulla*

③ SQL constraints are used to specify rules for data in a table.

1. NOT NULL column cannot have a null value.

(col1 int NOT NULL) * T33J72

2. UNIQUE all values in column are different.

(col2 int UNIQUE) * T9324I

3. FOREIGN KEY prevent actions that would destroy links between tables.

CREATE TABLE temp (

temp_id int (primary key),
FOREIGN KEY (cust_id) REFERENCES customer (id);

Null temp id blvntz 11. jst jmming. 6 jst 20 2021

temp

customer

cust_id
101
102

id
125
126

101 → 125
102 → 126

4. **DEFAULT** Sets the default values of a column.

Salary INT DEFAULT 25000;

5. **CHECK** It can limit the values allowed in a column.

CREATE TABLE table_name (
age INT CHECK (age >= 18));

① SELECT in detail:-

- ① used to select only data from the database.

→ Basic syntax.

SELECT col1, col2 FROM table_name;

- ① To select unique thing from table column:-

DISTINCT.

BETWEEN

⑥ WHERE clause:-

⑥ To define some conditions.

SELECT col1, col2 FROM table-name
WHERE condition;

⑦ Using operators in WHERE.

1. Arithmetic operators:-

+

(addition)

-

(subtraction)

*

(multiplication)

/

(division)

%

(modulus)

2. Comparison operators:-

=

equal to

!=

not equal to

>

>=

<

<=

3. Logical operators:-

AND

OR

IN

BETWEEN

ALL
LIKE (pattern)
ANY
OR

4. Bitwise operator :-
(pattern) 220

220 is of (1) 121 44

220 is of (1) 121 44

① Between :- . value of 2 is greater than 220

SELECT * FROM student WHERE marks BETWEEN 80 AND 90 ;

- (1) 121
- (1) 121
- (1) 121

① IN :-

SELECT * FROM student WHERE city IN ('Delhi', 'Mumbai')

SELECT * FROM student WHERE city IN ('Delhi', 'Mumbai')

① NOT :-

SELECT * FROM student WHERE city NOT IN ('Delhi', 'Mumbai')

SELECT * FROM student WHERE city NOT IN ('Delhi', 'Mumbai')

① ORDER BY clause

ORDER BY clause

① ORDER BY clause

SELECT * FROM student
ORDER BY marks ASC (ascending)

OR

DESC (descending)

→ Use (,) to separate.

① Aggregate functions:-

① Aggregate functions perform a calculation on a set of values and return a single value.

• COUNT

• MAX()

• MIN()

• SUM()

• AVG()

SELECT max(marks) FROM student;

SELECT avg(marks) FROM student;

② Group By Clause:-

① Group rows that have the same values into summary rows.

It collects data from multiple records and groups the result by one or more columns.

Generally we use group by with some aggregation function.

(same pattern as before) -> Example

→ Count number of student in each city.

SELECT city, count(name)
FROM student

GROUP BY city;

(same pattern as before) -> Example

① HAVING clause:-

① Similar to where i.e. applies some condition on rows. Used when we want to apply any condition after grouping.

(color) → Count number of students in each city where max marks cross 90

SELECT count(name), city FROM students
GROUP BY city.

HAVING max(marks) > 90;

② General order for report of a group or rows

marks and other info in some particular way

SELECT column(s) FROM table name

WHERE conditions.

GROUP BY column(s).

HAVING conditions

ORDER BY column(s) ASC or DESC

DESC;

② Table related queries group (small)

1. Update:- (to update existing rows).

UPDATE table_name.

SET col1 = Val1, col2 = Val2

WHERE conditions.

2. Delete:- (to delete existing rows)

DELETE FROM table_name.

WHERE condition

3. Restricting Foreign key:-

FOREIGN KEY (child table) REFERENCES table (parent table)

② Cascading Foreign Key

1. on delete cascade:-

② When we create a foreign key using this option, it deletes the referencing rows in the child table when the referenced row is deleted in the parent table which has a primary key.

2. on update cascade:-

② When we create a foreign key using update in the parent table which has a primary key.

Ex.

~~CREATE TABLE~~ CREATE TABLE student (course_id INT PRIMARY KEY,

course_id INT, FOREIGN KEY (course_id) REFERENCES course (id)

ON DELETE CASCADE ON UPDATE CASCADE);

~~TRUNCATE TABLE~~ TRUNCATE TABLE

① TABLE related query :- (name)

1. Alter :- (to change the schema)

Alter is used to modify the structure of the table. It is used to add, delete or modify columns, change data types, etc.

② ADD column

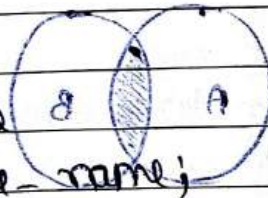
ALTER TABLE table name
ADD COLUMN col name datatype constraints;

③ DROP column

ALTER TABLE table name
DROP COLUMN column name;

④ RENAME table

ALTER TABLE table name
RENAME TO new table name;



⑤ CHANGE column (rename)

ALTER TABLE table name
CHANGE COLUMN old name new name datatype const.
RENAME COLUMN old name TO new name;
MODIFY COLUMN (modified datatype / constraint) :-

In Oracle

⑥

mysql 140

ALTER TABLE table_name
MODIFY col_name data_type constraint;

Table related queries

1. Truncate (to delete table's data)

TRUNCATE TABLE table_name;

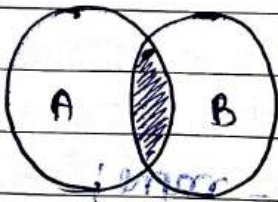
Joins in SQL :-

Join is used to combine rows from two or more tables, based on a related column between them.

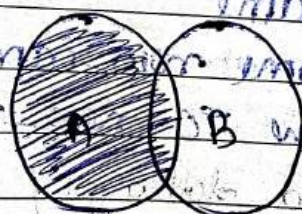
Types of joins

(Venn diagrams)

1. Inner Join :-

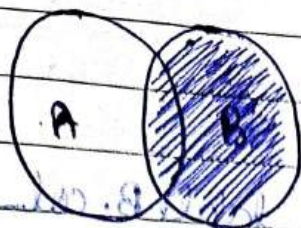


2. Outer join :-



left join

(ii)



Right Join

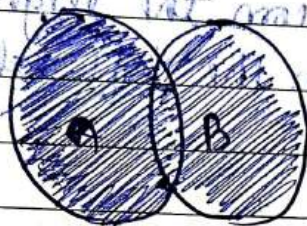
(2) example T31172

A id col. M097

B id col. M101 T31172

; more id. A id col. M10

(iii)



Full join

(2) example T31172

A id col. M097

B id col. M101 T31172

; more id. A id col. M10

① Inner Join:-

① Returns records that have matching values in both tables.

→ Syntax

SELECT column(s)

FROM table A

INNER JOIN table B

ON table A.col_name = table B.col_name;

; more id. A id col. M10

① Left Join:-

(2) example T31172

; more id. A id col. M10

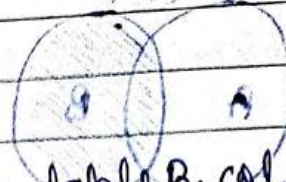
① Returns all records from the left table and the matched records from the right table.

; more id. A id col. M10

→ Syntax

; more id. A id col. M10

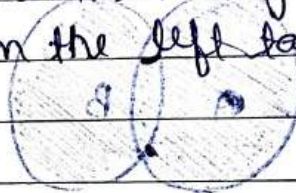
SELECT column(s)
FROM Table A
LEFT JOIN Table B
ON Table A.col_name = Table B.col_name;



not used

① Right Join:-

① Returns all records from the right table, and the matched records from the left table.



not used

→ Syntax

SELECT column(s)
FROM Table A
RIGHT JOIN Table B
ON Table A.col_name = Table B.col_name;

not used

not used

② Full Join:-

① Returns all records when there is a match in either left or right table?

(2) records 7.2.178

A record 1097

A record 1007 8.1.178

not used

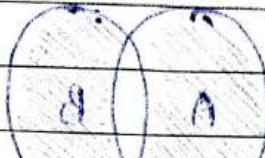
→ Syntax in MySQL:-
SELECT * FROM student
LEFT JOIN course
ON student.id = course.id

not used

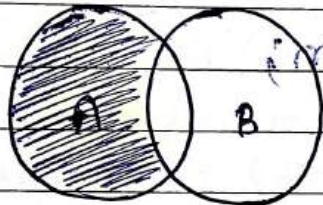
SELECT * FROM student
RIGHT JOIN course
ON student.id = course.id;

not used

→ Syntax for all joins:
 SELECT column(s)
 FROM table A
 FULL JOIN table B
 ON table A.col_name = table B.col_name.
 ∴ But not match value are null.



1. Left exclusive Join:-

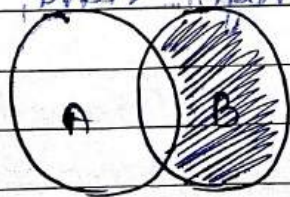


SELECT *

FROM student as s
 LEFT JOIN course as c
 ON s.id = c.id
 WHERE c.id IS NULL;

∴ (Student and course are two different tables)

2. Right exclusive Join:-



SELECT *

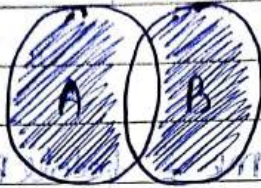
FROM table A

RIGHT JOIN table B

Intersection of course and student is not shown

ON table A, col name = table B, col name
WHERE table A, col name IS NULL;

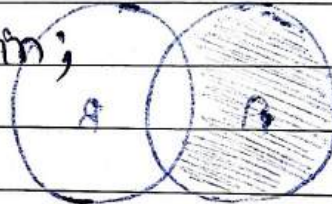
⑧



LEFT exclusive join

UNION ALL

RIGHT exclusive join;

⑨ Self Join:-

⑩ It is a regular join but the table is joined with itself.

```
SELECT column (s)
FROM table as a
JOIN table as b
ON a.col name = b.col name;
```

```
SELECT a.name as manager_name, b.name
```

to make
a new name
of new column



* 123456
Aldol 1409
Bldol 1409

① Union :-

① It is used to combine the result set of two or more select statements given unique records.

② To use it:-

- (i) every select should have same no. of columns
- (ii) columns must have similar data types.
- (iii) columns in every select should be in same order.

→

Syntax:-

2A (more data) W3IV 31A385

(more data) M097 31A385, 31A385, 31A385

SELECT column(s) FROM table A

UNION

SELECT column(s) FROM table B

① SQL Sub Queries:-

① A subquery or inner query or a nested query is a query within another SQL query.

It involves 2 select statements.

→

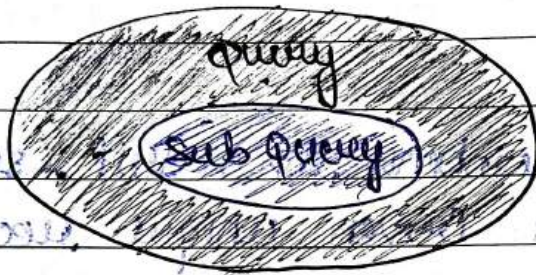
Syntax:-

SELECT column(s)

FROM table name

WHERE col name operator

(subquery);



① MySQL views

② A view is a virtual table based on the result set of an SQL statement.

→ Syntax:

CREATE VIEW (table-name) AS
SELECT col-name1, col-name2 FROM (Real table-name)

To see

SELECT * FROM (table-name)

③ Dual

④ View is a virtual table based on the result set of an SQL statement.